

docs/index.html · branch thecleaner · 3 freq bands: **BASS** 0-15 · **MID** 40-120 · **TREBLE** 150-300 · **ENERGY** = (B+M+T)/3
 Legenda stato: **ON** attivo · **OFF** disattivato (flag) · ~ modulato da calmness in silenzio · riga azzurra = aggiunta su thecleaner

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Stato	Trigger	Dwell	Effetto
silenceMode	energy < 0.08	2 s	respiro cerchio, blue tint, stelle dim · alimenta calmness
calmness 0-1	spring-damped → silenceMode	~0.8 s ramp	attenua: stelle, hex, spirale, vortex, glow, tunnel · body.silent ferma CSS
raveMode	energy > 0.65	5 s	tint caldo, glow shadow del cerchio espanso
marathonMode	playback > 15 min	—	1 stella su 5 diventa "warm" (sprite warm)
dropDetector	energy avg <0.15 + bass spike >0.7	cd 5 s	flash bianco, 3 pulse rings, sacred geo boost
BPM stable	4+ beat con varianza < 100ms²	—	spirale + sacred geo si sincronizzano
pupilDilation 0-1	beat steady (deviation < 0.3)	20 beat max	cerchio centrale si dilata
psychState	cambio ogni 30-60 s random	varia	6 stati: normal / trip / khole / strobe / aberration / vhs

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drawProceduralBg() · render-frequency desktop=1, mobile=2, slow=4
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#	Layer	Trigger	Soglia / mapping	Stato
L0	Star Field / Warehouse Dust (200 particles)	TREBLE twinkle + BPM boost	$\alpha = 0.25 + \text{treble} \cdot \text{twinkle} \cdot 0.5 + \text{bpmPulse} \cdot 0.4$	ON ~drift halts + dim
L0b	Constellation Lines (hi-hat onsets)	TREBLE onset (vs baseline $\times 1.55$)	$\text{treble} > 0.22 \ \& \ \text{ratio} \cdot 1.55 \ \& \ \text{cd} \ 140 \ \text{ms}$	ON ~suppress
L0.5	Radial BPM Wave (cerchio espandente)	BPM stable	$\text{bpmPulseIntensity} > 0.05$	ON
L1	Pulse Rings (concentric, da centro)	BASS peaks $\cdot$ drop spawn $\times 3$	$\text{bass} > 0.5, \ \text{cd} \ 300 \ \text{ms} \cdot \text{drop: } 3 \ \text{ring simultanei}$	ON
L2	Hex Grid (pre-rendered, blitted)	BASS + BPM boost	$\alpha = 0.6 + \text{bass} \cdot 1.0 + \text{bpm} \cdot 0.3$	ON ~fade
L3	Spiral Vortex (14 arms, radial gradient)	BASS amplitude $\cdot$ BPM sync rotation	$\alpha = (0.15 + \text{bass} \cdot 0.2) \cdot \text{activity} \cdot \text{radial gradient} \cdot \text{width} \ 1.5 + \text{bass} \cdot 2$	ON ~fade $\cdot$ post-fix: 14 stroke/frame (era 1120)
L5	Central Circle / Pupil (broadcast icon)	BASS scale + dilation + breath	$r = \text{minDim} \cdot 0.06 \cdot (1 + \text{bass} \cdot 0.1 + \text{pupil} \cdot 0.5) \cdot \text{silenzio: breath} \ \sin(t \cdot 0.002)$	ON ~glow tighter
L6	Sacred Geometry (Flower of Life)	ENERGY $> 0.6 +$ BPM sync	$\alpha \ \text{target} = 1 \ \text{if} \ \text{energy} > 0.6 \ \text{else} \ 0 \cdot \text{drop boost}$	ON
L8	State Overlays (tint)	drop flash $\cdot$ rave tint $\cdot$ silence blue	drop: white $\alpha \ 0.3 \cdot$ rave: amber $\alpha \ 0.04 \cdot$ silence: blue $\alpha \ 0.05$	ON

Layer	Trigger	Cost note	Stato
Concentric Rings rotanti	ENERGY rotation · BASS alpha	shadowBlur=15 sempre (target tier P0.2)	ON ~clear forzato sopra calmness 0.85

Budget per beat: **FLASH** max 1 · **ACCENT** max 3 desk / 4 mob · **DISTORT** max 2 desk / 3 mob

Triggerati da BASS

Effetto	Soglia	Stato
crowdStrobe (unconditional)	ogni beat	ON
spatialStrobo (5 pattern)	bass > 0.8 · FLASH	ON
spiralVortex bass (5 elementi)	bass > 0.8 · ACCENT	ON
zoom-burst	bass > 0.8 + cd 600 ms · DISTORT	ON
shake-burst	bass > 0.9 + cd 1.5 s · DISTORT	ON
strobeFlash (bianco full-screen)	bass > 0.85 · FLASH	ON
VHS lines spike	bass > 0.8 · DISTORT	ON
negativeLayer rays (single/multi)	bass > 0.7 · DISTORT	ON
fractalTear (6 clip-path: octagon, star, etc)	bass > 0.78 · ACCENT	OFF (l'invasore)
circularStrobo (4 anelli da 50→150vw)	bass > 0.8 · ACCENT	ON
distortZone (3 zone speaker stack)	bass > 0.75 · DISTORT	ON
geo-strobe cerchi/radial/bottom	bass > 0.75 · ACCENT	ON

Triggerati da MID

Effetto	Soglia	Stato
spiralVortex mid (implode)	mid > 0.75 · ACCENT	ON
silverGlowParticle (8 elementi)	mid > 0.6 · ACCENT	ON
spiralWhirl (4 SVG spirali rotanti)	mid > 0.65 · ACCENT	ON
shimmerTwinkle (5 puntini scale 1→3)	mid > 0.5 · ACCENT	ON
warpZone (antenna + crowd)	mid > 0.7 · DISTORT	ON
strobeLeftWall / RightWall	mid > 0.7 (fuori budget)	ON
speaker-flash laterali (always-on)	$\alpha = \text{mid} \cdot 1.2$	ON
dreamLayer (4 immagini blurred)	bass > 0.6 + mid > 0.7 · rare	ON

State Layer (psychState)

Stato	Peso	Effetto	Durata
normal	30%	baseline	30–60 s
trip	20%	breath 4s loop su #background	15–30 s
khole	15%	blur 3px + contrast 1.5 + saturate 0 (B&W)	10–20 s
strobe	10%	10–30 flash ravvicinati	5 s
aberration	15%	drop-shadow chromatic shift	8–15 s
vhs	10%	vhs lines α 0.6 + grain α 0.4 permanenti	10–20 s

Blur & Movement (coalesced)

Sistema	Sorgenti	Smoothing
currentBlur (su #background)	burstBlur (zoom-burst) + midRamp (mid > 0.6 ramp soft fino a 0.8)	spring-damper $k=0.05$ / $d=0.78 \cdot \text{target}$ / decay easing
transform main	zoomIntensity + depthZoom + rotateAmount + sway X/Y + rotateZ/X + burst offset	POV swirl, BPM-cyclic 2/3/2.5 s

Triggerati da TREBLE

Effetto	Soglia	Stato
spiralVortex treble (pulse)	treble > 0.7 · ACCENT	ON
negativeLayer treble (spiral/cross)	treble > 0.65 · DISTORT	ON
circularStrobo treble (2-3 in cascata)	treble > 0.75 · ACCENT	ON
chromaZone (zone laterali)	treble > 0.7 · DISTORT	ON
strobeTop (antenna beam)	treble > 0.75 (fuori budget)	ON

Triggerati da ENERGY / altri

Effetto	Soglia	Stato
grain spike	energy > 0.7 · DISTORT	ON
geo-strobe random	energy > 0.8 (fuori budget)	ON
circlePulse (DOM halo)	$\alpha = (0.5 + \text{bass} \cdot 0.5) \cdot \text{activity}$	ON ~dim 25%

Continui (CSS animations)

Effetto	Trigger	Stato
tunnelEffect rings (50→2000px, 3s loop)	sempre	ON ~opacity = activity · target P0.3: transform scale
dust-particles (15 punti, 20s)	sempre	ON · pausa con body.silent
vhs-line (5 linee, 8s scan)	sempre + state	ON · pausa con body.silent
grainOverlay (0.5s steps)	sempre	ON · pausa con body.silent

micro-movements	5%/frame random jitter	skip se calmness > 0.4
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Performance (cosa già fatto / cosa resta)

#	Fix	Stato
P0.1	Spirale 1120 → 14 stroke/frame (radial gradient)	DONE
P0.7	Star sprite glow (no fillRect duro)	DONE
P2.1	Beat budget (Flash×1 / Accent×3 / Distort×2)	DONE
Blur	Spring-damper sul filter blur	DONE
Silence	calmness 0–1 + body.silent + fade su tutte le sorgenti	DONE
L0b	Constellation lines su hi-hat onsets	DONE
P0.2	Vortex shadowBlur=15 → gate sotto bass > 0.5	TODO
P0.3	tunnel-zoom da width/height → transform scale	TODO
P0.4	circle-pulse-expand → transform scale	TODO
P0.5	circlePulse box-shadow 3-layer → 1-layer	TODO
P1.1/2	will-change su tutti gli overlay animati	TODO
P5.1	document.hidden pause animate()	TODO
P5.2	frame-rate downshift su calmness > 0.85	TODO
Asset	Step A: 24 file orfani rimossi (12 MB)	STAGED

Aperti / da decidere

- **Mask pupilla per i raggi** — domanda da 1M\$, via mask: radial-gradient() con CSS var dinamica legata al raggio del cerchio centrale.
- **Chicche per momenti senza cassa (B&W autoriale)** — già scelta #1 = constellation lines (DONE). Prossime in coda: inverted-flash su snare · echo from the past su drop · scan-line scrub su mid sostenuto.
- **"Stelle enormi quadrati"** = fractal-tear → disattivati (FRACTAL_TEAR_ENABLED = false). Da decidere se rimuoverli definitivamente o rifare il look.